

Week 2: Data Cleaning Log & Quality Improvement Report

Dataset: DePaul Dataset

Team: SLU 0203 DVT Team 3

Prepared by: Abiodun Oluwafemi Toyosi

Clifford Oduor

Joshua Anayochukwu

Kelechi Emmanuel

Sakshi Doijad

Tanmay Sawant

Tasnia Sultana

Temiloluwa Bolaji

Toluwalope Sokefun

Saminu Mustapha

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Data Cleaning Log

Data cleaning is important for ensuring the dataset is accurate and reliable for analysis. The dataset used contains 7,543 records and 80 variables. During the cleaning process, issues such as missing values, duplicate columns, inconsistent formatting, and incorrect data types were identified and corrected.

Handling Missing Data

Some columns contained missing values, and a few columns were completely empty. Columns with 100% missing values were removed because they did not contribute useful information. In some cases, missing categorical values were replaced with “Unknown” while meaningful missing values were kept.

Identifying and Removing Duplicates

Duplicate columns were found in the dataset due to system export issues. For example, some variables appeared twice with similar names. These duplicates were removed to simplify the dataset and avoid redundancy.

Correcting Inconsistencies

Text fields had inconsistent formatting such as differences in capitalization (e.g., *india* vs *India*). These were standardized to ensure consistency and accurate grouping during analysis.

An overview on the details of the data cleaning process was done can be accessed here:

 DePaul_Data_Cleaning_Process

https://docs.google.com/document/d/1eCrLccrV3XWw2Gj08vhT_iGNomOZhAR8u7eupq6UJ08/edit?usp=sharing

While the cleaned dataset can be accessed here:

 De Pauls Cleaned Dataset

https://docs.google.com/spreadsheets/d/1bX_4lvikvudR1uAEJqnQRJD8ajIJyAQECmu5szk9I4/edit?usp=sharing

Fixing Data Types and Formats

Some fields had incorrect formats, particularly date and phone number fields. Dates stored as text were converted into proper date formats, and phone numbers were stored as text to preserve leading zeros.

Before vs After Data Comparison

Before cleaning, the dataset contained several quality issues such as missing values, duplicated columns, and inconsistent formatting. After the cleaning process:

- Empty columns were removed
- Duplicate columns were eliminated
- Text values were standardized
- Date formats were corrected

These improvements significantly increased the accuracy and reliability of the dataset.

EDA Insights Report (Analysis Section)

This exploratory data analysis examines outreach and application data to identify trends in engagement, program demand, geography, and application sources.

Outreach Coverage:

Most applicants have not been contacted. About 64% are “Not Contacted” while 36% have been contacted, indicating opportunities to expand outreach.

Outreach Attempts:

Around 64% of records show zero outreach attempts, suggesting that increasing follow-ups could improve engagement.

Program Demand:

Business Analytics and Computer Science have the highest number of applicants, followed by Data Science, Health Informatics, and Cybersecurity, showing strong interest in technology-related programs.

Geographic Distribution:

Hyderabad has the highest number of applicants, while many records fall under “Unknown”, indicating incomplete location data.

Most applications (84%) come from Non-Study Group sources, while 16% come from Study Group channels.

Here is what was done ;

https://docs.google.com/document/d/1AchHZkzIkvwccDJO_brA75T2PaUOJ5u6TZWJmjtcF-E/edit?usp=sharing

After cleaning, the dataset became more consistent, structured, and reliable. These improvements ensure the data is ready for accurate analysis and visualization in the next stage of the project.

A draft dashboard was designed using Microsoft Power BI to visualise some of the most important outreach KPIs and highlight actionable insights in accordance with the required deliverables.

The KPIs (outreach volume, response rate and conversion rate) were calculated using the Measure function in Power BI, and then visualized as Cards at the top of the dashboard. The Outreach Volume was 5000, the Response Rate was 36.14%, while the Conversion Rate was 29.83%.

The other visualizations that were included are, Outreach Volume by Region, Response Rate by Outreach Action, Conversion Rate by Program, Outreach Trend over Time and Outreach Channel Effectiveness by Outcome. Two slicers (Region and Program) were also included to facilitate easy navigation through the information on the dashboard.

From the dashboard, it was deduced that the region with the highest outreach volume was **Telangana (724)**, **Sent for Evaluation (34.93%)** dominated the response rate of the Outreach Actions, **April of 2024 (149)** recorded the highest Outreach Volume, and over half of the students (**50.22%**) were not assigned to a study group due to **Missing Transcripts**.

<https://app.powerbi.com/view?r=eyJrIjoib2M1MjIjYWQYzE2ZS00OWM5LTlmMjktNTc2YWVhOGI3ZiE1IiwidCI6ImFmYm01YzZM3LTvjMmUtNGYxNS1hY2FmLTU5ZTYxZWZzNTc5ZiJ9>